Distributed Computing + AI/Data Science Applications (5-Minute Revision Sheet)

Fundamentals of Distributed Computing

• Distributed System: Multiple computers (nodes) connected via a network working together as one system.

• Key Characteristics: Resource sharing, concurrency, scalability, fault tolerance, transparency, heterogeneity.

• Goals: Resource sharing, reliability, scalability, performance, openness.

• Types: Distributed computing systems (clusters/cloud), distributed information systems (databases), distributed pervasive systems (IoT).

• System Models:

  - Physical Model: Hardware nodes and network connections.

  - Architectural Model: Client–Server, Peer-to-Peer, Multi-tier.

  - Fundamental Model: Interaction (communication), Failure (crashes, delays), Security (authentication, encryption).

AI & Data Science in Distributed Systems

• Used when datasets and computations are too large for a single machine.

• Core ideas:

  - Distributing computational tasks across nodes.

  - Handling large volumes of data using distributed storage.

  - Leveraging parallel processing for faster computation.

• Benefits: Faster training, scalable analytics, real-time processing.

Key Issues in Distributed Systems

• Data Storage & Retrieval: Data distributed across nodes; requires partitioning, indexing, replication.

• Data Consistency: Ensuring all nodes see the same updated data.

- Strong Consistency: Immediate updates everywhere.
- Eventual Consistency: Updates propagate gradually.

• Communication Overhead: Network delays, message passing, data transfer.

• Synchronization: Coordination between processes (logical clocks, mutual exclusion).

• Fault Tolerance: System continues working despite node failures using replication, redundancy, checkpoints.

Major AI + Distributed Computing Applications

Predictive Maintenance

• Uses sensor data + AI to predict equipment failures.

• Benefits: Reduced downtime, lower maintenance cost.

Fraud Detection

• Detects abnormal financial transactions using ML models.

• Used in banking, payments, insurance.

Intelligent Transportation Systems

• AI analyzes traffic data for congestion prediction, route optimization, accident detection.

Supply Chain Optimization

• Predict demand, manage inventory, optimize logistics and delivery routes.

Energy Management

• Smart grids monitor energy usage, forecast demand, integrate renewable energy.

Healthcare & Medical Diagnostics

• AI analyzes medical images, patient records, wearable data.

• Supports disease diagnosis, telemedicine, remote monitoring.

Customer Behavior Analysis

• Studies purchasing patterns for marketing and recommendation systems.

Natural Language Processing (NLP)

• Enables machines to understand human language.

• Applications: chatbots, translation, sentiment analysis, document classification.